CITIES OF HADITH

Quantifying Transmission in the First 400 Years of Islam

Abstract

Scholarship on the history of hadith transmission typically relies on two types of information: the documentation of transmission activity in the hundreds of thousands of isnads of hadiths, and the biographical data on tens of thousands of individual transmitters. Most studies rely on one type to the exclusion of the other. By relying on the largest digitized collection of Sunni hadith texts (Gawamiʿ al-Kalim), our paper combines the information in the two datasets to precisely quantify transmission activity in the different cities of the early Islamic empires during the first 400 years of Islam. We examine the cities with the most transmission activity in different times: Medina (0-119), Kufa/Basra (120-159), Kufa (160-193), Basra (194-227), and finally Baghdad (228-400). We show that the rise and fall of transmission activity in each of these cities correlates with the rise and fall of their demographic, economic, and political importance.

Keywords: Islam, Hadith, Digital Humanities

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Introduction

Scholars in the field of hadith studies engage in two fundamental types of analysis: *local* and *global*. Those doing a *local* analysis study the variations in the texts of a small set of hadiths and the structure of their isnads that purport to transmit them. The latter involves identifying the cities in which the narrators transmitted and the years in which they lived. While the isnads and texts are found in hadith collections, information about the time and cities in which narrators lived and narrated hadith are found in the works of narrator biography and criticism. Thus, scholars seeking to do a historical analysis of hadiths combine two types of data: the information found in hadiths' isnads documenting their purported transmission history with the time and place information found in biographies of individual transmitters. The results of these studies are of two types. One type dates and provides a geographical location for the transmission of a hadiths that exemplify a particular theme or idea. These results are often supplemented by an analysis of the context that explains why those hadith emerged and proliferated when and where they did. In these studies, the method is a close reading of a few texts and resultant conclusions apply only to them or a limited theme or idea. The second type of local analysis attempts to make far-reaching conclusions about the nature of hadith transmission overall gleaned from a study of a handful of hadith texts. For example, based on anachronisms in a few hadiths generated by later political and sectarian conflict, Goldziher concluded that an analysis of isnads could not yield historically accurate conclusions about the date of hadiths' circulation and regarded most of them as unauthenticatable through the methods developed by premodern hadith scholars. Similarly, Schacht, through an analysis of a few hadiths that evinced growth in the early part of isnads, concluded that this is a general feature of all legal hadiths.[1]

[1] For a useful survey documenting the range of approaches to hadith studies in the Western academy, see Harald Motzki, "Dating Muslim Traditions: a Survey," *Arabica* 52, no. 2 (2005), especially his characterizations of Goldziher and Schacht's analysis and conclusions.

Global studies, which are substantially less dominant in the field, are devoted to tracing the history of the practices of hadith transmission over a long period of time and wide geographic expanse, by relying only on the information found in narrators' biographies. These sources contain not just information on individual transmitters but also anecdotes about transmission practices and the motivations of various agents involved in hadith transmission. Both the quantity and type of information found in these sources makes them conducive to studies that take a *macroscopic* view of change and development in hadith transmission, because they are not limited to a few hadith texts or a particular theme.[2] Due to the magnitude and structure of the data, many of the arguments and conclusions found in these types of study take a statistical form. The conclusions made about nature of hadith transmission that result from studies of this type are on surer epistemic ground because they rely on a much larger dataset. But, unlike local analysis, which combines information found in hadiths and their isnads with data found in narrator biographies, global studies rely only on the latter.

The most conceptually and temporally ambitious of the global type of analysis is Regep Senturk's *Narrative Social Structure: Anatomy of the Hadith Transmission Network, 610-1505*.[3] Senturk's study attempts to provide a macroscopic view of the practices and individuals involved in hadith transmission over almost a millennium. He calls the set of intellectual and social practices involving thousands of individual hadith scholars, and the many prominent cities of Muslim empires and kingdoms that constituted and sustained hadith transmission over many

[2] For examples of such works, see: Asma Sayeed, *Women and the transmission of religious knowledge in Islam* (Cambridge: Cambridge University Press, 2013), who analyzes narrator biographies in order to produce a history of women's hadith transmission activities; Judith Ahola, "The community of scholars: an analysis of the biographical data from the Ta'rīkh Baghdād" (Ph.D., University of St Andrews, 2005) who looks at the social history of the community of hadith scholars in Baghdad over 320 years based on information recorded in al-Khaṭib al-Baghdadi's *Taʾrikh Baghdad*; and Scott C. Lucas, *Constructive critics, Ḥadīth literature, and the articulation of Sunnī Islam: the legacy of the generation of Ibn Sa'd, Ibn Ma'īn, and Ibn Ḥanbal*, Islamic history and civilization. (Leiden: Brill, 2004), who uses the analysis of narrator criticism and biography sources to make conclusions about the formation of Sunnism in early Islam.

[3] Recep Şentürk, *Narrative social structure: anatomy of the Hadith transmission network, 610-1505* (Stanford: Stanford University Press, 2005).

centuries, the *hadith transmission network*. Senturk shows how the coherence and temporal continuity of the hadith transmission network, with the amount of data recorded about it in hadith literature, allows one to analyze its features, not just qualitatively, but also quantitatively.

Our paper adopts and develops Senturk's conceptualization of hadith transmission as best explained as a diachronic social network devoted to the preservation, transmission, and criticism of hadiths. But, whereas Senturk relied only on information found in narrator biographies, we combine this biographical data with the information documenting transmission in 632,869 isnads taken from 1348 collections of hadith. We also combine all of the transmitter relationships from the isnads into a unified social network, allowing us to use existing mathematical and computational methods of analysis for historical interpretation.[4]

We demonstrate the power of global analysis that draws on data from ***both*** hadiths and narrator biographies by looking at one feature of hadith transmission that has received scant systematic attention: how the amount of transmission activity in different cities changed over time.[5] Until recently, it would have been unfeasible even for a team, let alone an individual

[4] For an introduction to social network analysis, see John Scott, "Social Network Analysis," *Sociology* 22, no. 1 (1988).

[5] Senturk is the only scholar we know of to systematically document transmission activity over time and place. He relies on the temporal and geographical information found on hadith narrators in a single collection of hadith biographies. See Şentürk, *Narrative social structure: anatomy of the Hadith transmission network, 610-1505*, 40 for a chart documenting presence of hadith transmitters in different locales in different time periods. Şentürk relies on Ibn Ḥibban's work, *Mashahir al-ʿulamaʾ al-amṣar wa aʿlam fuqahaʾ al-aqṭar*. For his description of the work see ibid., 41. For the way he defines the temporal layers containing the hadith transmitters, see ibid., 39-40. Most studies of the development of hadith literature over time use a combination of information drawn from anecdotes on various specific personalities involved in hadith transmission found in biographical sources with an intuitive qualitative judgement on the number of isnads important individuals are found in to describe the growth and evolution of hadith literature in the first three two four centuries of Islam. For a view that summarizes this scholarship, see Pavlovitch Pavel, "Ḥadīth", Encyclopaedia of Islam, Three and Jonathan Brown, *Hadith: Muhammad's legacy in the medieval and modern world* (Oxford: Oneworld, 2009), 15-63. They track the change in the mode of transmission from largely oral and informal to formal, and the role of state patronage and the work particular individuals, such as Ibn Shihab al-Zuhri, in the process. An earlier scholar, Aʿẓami notes that as more individuals became involved in hadith transmission, the number of isnads documenting transmission grew to an unwieldy amount, "from a few thousand aḥadith" to "three quarter of a million in the mid-third century". He cites Ibn Ḥanbal as estimating 7,000,000 hadith in existence, and al-Bukhari as claiming he had collected 600,000. This is the only attempt at quantifying hadith transmission activity that we found in existing scholarship. Muḥammad Muṣtafā Aʿẓamī, *Studies in*

scholar, to precisely quantify and analyze the geographical and temporal development of hadith transmission activity. Doing so would have required mapping the millions of names cited in the isnads of hadiths to the biographical information associated with them found in thousands of narrator biographies.[6] But the recent digitization of books of hadith, biographies of narrators, and the linking of the narrators found in the hadiths to information about them in biographies has now made this possible. The largest such enterprise for Sunni collections was published in the form of a freely available downloadable software package called *Gawamiʿ al-Kalim* (*GK*).[7] We restructured, synthesized, and supplemented *GK*'s data; we created a social network of transmitters; and we used the network to quantify the transmission of hadith in the major cities of early Islam during its first 400 years.

We discovered that the largest proportion of hadith activity in the period under consideration is located in four cities: Medina, Basra, Kufa, and Baghdad. From the year 0 to 119, Medina was the dominant city.[8] Within that period, Kufa and Basra start emerging as major centers in the year 80. Kufa decisively replaces Medina as the top city in 120. From 120 to 160, Kufa and Basra traded places as the top city. From 160 to 193 Kufa and Basra were the dominant cities. Basra is the dominant city from 194 to 227. Baghdad starts emerging as a major center in 185 and overtakes Basra in 230. From then until 400, it is the dominant city, with exceptions in a few years. The level of precision with which we are quantitatively able to document the amount of hadith transmission in each of these cities was previously impossible.

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early hadīth literature: with a critical edition of some early texts (Indianapolis: American Trust Publications, 1978), 199, 301-04. One may find a scattering of casual observations that note the rise and decline of different cities as centers for hadith transmission throughout Islamic studies scholarship. For example, in an encyclopedia article on the city Kufa, Meir Litvak notes that Kufa was a leading center of hadith transmission in the seventh century. See Meir Litvak, "KUFA", Encyclopædia Iranica. Similarly, Scott Lucas notes that political instability caused hadith transmission activity to move from Kufa and Basra to Baghdad in the 3rd/9th century. See Lucas, *Constructive Critics*, 73.

[6] For a recent paper introducing machine learning techniques to automatically identify narrators in isnads based on GK's corpus, see Mohamed Alkaoud and Mairaj Syed, "Learning to Identify Narrators in Classical Arabic Texts," *Procedia Computer Science* 189 (2021).

[7] For a description of the type of hadith data in *Gawamiʿ al-Kalim*, see Mairaj Syed et al., "Verifying Source Citations in the Hadith Literature," *Journal of Medieval Worlds* 1, no. 3 (2019): 8-10

[8] We use the Hijri calendar dating system throughout this paper.

As we will show, the years in which these cities rose and fell, and their sequence, correlates with their demographic, economic, political, and intellectual fortunes. This correspondence is a strong argument against those who are globally skeptical of the historicity of all of the isnads of hadiths.

In addition to making substantive historical claims about the quantity of transmission activity in the hadith network, our paper also makes two methodological innovations of broad applicability for the fields of hadith and the history of early Islam. Whereas previous global studies of hadith transmission simply counted the number of transmitters in a given city and year according to information found only in the biographical dictionaries, we take the transmitter relationships that make up the isnads as basic unit of our analysis. Doing so, has two advantages. First, since we are trying to measure transmission activity, an instance of transmission requires not just the presence of a single individual in a particular city and year, but the two narrators composing a relationship. Thus, the transmitter relationship, as opposed to the individual narrator, is more appropriate. Second, because in many cases the temporal and geographic data associated with transmitters is obtained from different biographical sources, placing a relationship in a given year and city potentially synthesizes information from different textual streams, thereby allowing us to corroborate geographic and temporal information gleaned from the biographies of different individuals. But, placing a transmitter relationship in different years and cities required developing methods to extend year and city information associated with individual narrators to the transmitter relationship itself. We do so by transforming the years in which the source and recipient died (and when available, when they were born) into a time span in which their lives overlapped and in which transmission between them *could* have taken place. Similarly, we transform the often-multiple cities the biographies records as places in which narrators of a given transmitter relationship lived or visited into the *possible* cities a transmission *could* have taken place.

Description of Data Set

GK performed two valuable operations that make the calculation of the amount of hadith transmission activity in different cities and years possible.[9] They identified the unique narrators present in the isnads of Prophetic hadiths and linked them to a table containing 19 types of discrete information for each narrator, which they mined from medieval biographical sources. We demonstrate how *GK* combined and structured this data by looking at the famous five pillars hadith:

> **Al-Marwazi/ID:30844** [hadith collector]: "**Saʿid b. Masʿud/ID:17768** narrated to me, **Ishaq b. Mansur/ID:969** narrated to us, **Israʾil/ID: 982** narrated to us, from **Jabir/ID:2075**, from **al-Shaʿbi/ID: 4099**, from **Jarir b. ʿAbdullah/ID:2123**, who said the **Prophet/ID: 99999**, may peace and blessings be upon him, said: Islam is built on five [pillars]…"

Except for the Prophet Muhammad, which we added in ourselves (ID 99999), *GK* has labelled each of the narrators in the isnad and assigned them a unique ID. The unique ID for each of the narrators corresponds to a row that contains biographical information for each narrator. See table 1 below for narrators from the five pillars hadith.

ID	Famous Name	Birth Year	Death Year	Cities
99999	Prophet Muhammad		11	Medina
2123	Jarir b. ʿAbdullah		51	Kufa, Qirqisiyya, Yemen
4099	al-Shaʿbi	20	105	Kufa
2075	Jabir b. Yazid		128	Kufa
982	Israʾil b. Yunus	100	160	Baghdad, Kufa
969	Ishaq b. Mansur		204	Kufa
17768	Saʿid b. Masʿud		271	Marw
30844	Muhammad b. Nasr	202	294	Nisabur, Khurasan, Marw

Table 1. Example of GK Narrator Biography Data

[9] Syed et al., "Verifying Source Citations in the Hadith Literature,".

Since we use the transmitter relationships found in the isnads as the fundamental unit for calculating the quantity of hadith transmission, let's look at just the list of transmitter relationships from the five pillars hadith:

1. **Prophet Muhammad**/ID 99999 → **Jarir b. ʿAbdullah/ID:2123**
2. **Jarir b. ʿAbdullah/**ID: 2123 → **al-Shaʿbi/**ID: 4099
3. **al-Shaʿbi/ID: 4099** → **Jabir/ID: 2075**
4. **Jabir b. Yazid**/ID: 2075 → **Israʾil/ID: 982**
5. **Israʾil b. Yunus**/ID:982 → **Ishaq b. Mansur/ID:962**
6. **Ishaq b. Mansur**/ID:969 → **Saʿid b. Masʿud**/ID:17768
   Saʿid b. Masʿud/ID:17768 →**Muhammad b. Nasr Al-Marwazi**/ID: 30844

When we add in the city and year information from the biography table, we see that there are quite a few ways in which a transmitter relationship can be distributed in time and space, and these ambiguities need to be systematically addressed before one may calculate hadith transmission activity (as seen in table 2).

Source	Source Birth Year	Source Death Year	Source Cities	Recipient	Recipient Birth Year	Recipient Death Year	Recipient Cities
Prophet/ID 99999		11	Medina	**Jarir b. ʿAbdullah/ ID:2123**		51	Kufa, Qirqisiya, Yemen
Jarir b. ʿAbdullah/ID:2123		51	Kufa, Qirqisiya, Yemen	**al-Shaʿbi/ID: 4099**	20	105	Kufa
al-Shaʿbi/ID: 4099	20	105	Kufa	**Jabir/ID: 2075**		128	Kufa
Jabir/ID: 2075		128	Kufa	**Israʾil/ID: 982**	100	160	Baghdad, Kufa
Israʾil/ID: 982	100	160	Baghdad, Kufa	**Ishaq b. Mansur/ID :962**		204	Kufa
Ishaq b. Mansur/ID: 962		204	Kufa	**Saʿid b. Masʿud**/ID :17768		271	Marw
Saʿid b. Masʿud/ID: 17768		271	Marw	Al-Marwazi/ID : 30844	202	294	Nisabur, Khurasan, Marw

Table 2. Years and Cities of Transmitter Relationships

One ambiguity deals with selecting the city or cities in which we should place the transmitter relationship. This is not a problem for the **al-Shaʿbi**/ID: 4099 → **Jabir**/ID: 2075 because both narrators belong to Kufa. But what about when the transmitters of a relationship belong to different cities as is the case with the **Ishaq b. Mansur**/ID:969 → **Saʿid b. Masʿud**/ID:17768 relationship? That is, Ishaq belongs to Kufa and Saʿid belongs to Marw. Or what about when both transmitters in a relationship belong to multiple cities and have more than one city in common? The other fundamental ambiguity deals with identifying the years in which to place the transmitter relationship. In the best-case scenario, we have the years in which the transmitters of a relationship were born *and* dead. We do not, however, have the data on the precise years in which the transmitters could have met. Furthermore, in most cases, all we have is the death years for the narrators. We, therefore, devised two algorithms to inferentially resolve these ambiguities in a way that allows us to place transmissions in *possible* cities and years (see appendix 1 for details).

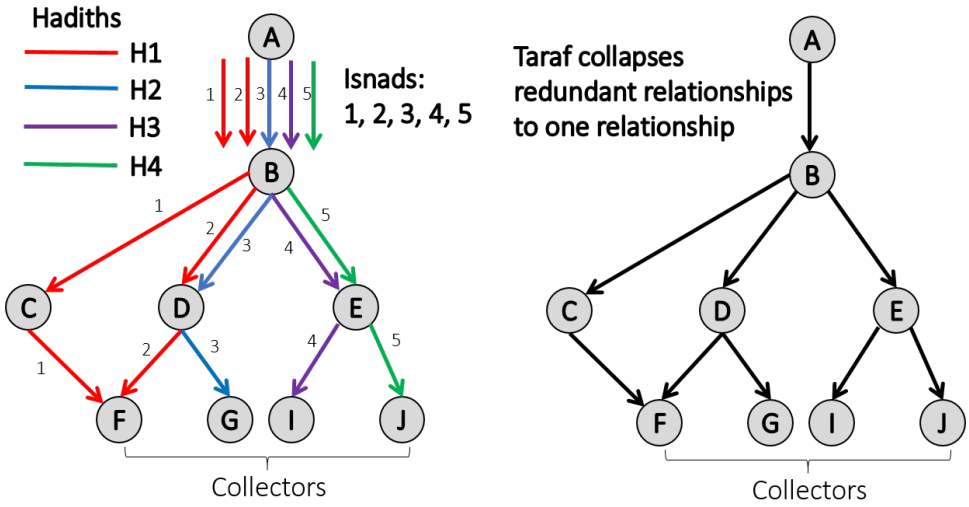

Figure 1. Distinguishing Isnad (denoted by numbers), Hadiths (denoted by colors), and Taraf (denoted by black arrows).

The last ambiguity we had to resolve is exactly what we mean by hadiths for the purpose of quantifying them. Based on the structure of *GK*'s dataset, we faced one of three possibilities. The simplest is when a hadith consists of one isnad and one text. Of *GK*'s 447,205 prophetic

hadiths, 342,255 have this form. But the rest have a one-to-many relationship between hadiths and isnads because it is often the case that a single hadith provides multiple isnads for the same text. Finally, many different collections of hadith contain versions of the same text. In that case, we have a many-to-many relationship between isnads and hadiths. *GK* has helpfully identified these hadiths across different hadith collections by tagging them with a unique numerical identifier, which we dubbed the *taraf* ID. Thus, *GK* reduced the 447,205 hadith texts to 20,744 unique *taraf* IDs, which comprise different versions of the texts. For example, *GK* assigned the five pillars hadith the unique ID 523 and tagged all of the five pillars hadith across 80 different hadith collections in which it is found with this ID. Since we calculate the amount of transmission activity in any given year and city by counting the number of times a transmitter relationship is cited in a text, we had one of three different ways of counting it: the number of times it occurs in distinct isnads (as indicated by the numbered arrows in figure 1), the number of times it occurs in distinct hadith texts (as indicated by the color of the arrows in figure 1), and the number of times it occurs in distinct *tarafs* (as indicated by the black arrows of the isnad diagram on the right in figure 1). For example, in the *GK* dataset, the **Jarir b. ʿAbdullah/**ID: 2123 → **al-Shaʿbi/**ID: 4099 transmitter relationship occurs in 320 isnads, 214 hadiths, and 11 tarafs. In the five pillars taraf alone, this relationship occurs in 36 isnads and 27 hadiths. By comparison, because the **Jarir b. ʿAbdullah/**ID: 2123 → **al-Shaʿbi/**ID: 4099 is a relatively early transmission relationship (occurring immediately after the Prophet) it occurs in the isnads of many hadith texts ultimately recorded in many different hadith collections. The recurrence of the **Jarir b. ʿAbdullah/**ID: 2123 → **al-Shaʿbi/**ID: 4099 transmitter relationship generally matches the structure of hadith transmission overall: early authorities and transmitter relationships occur much more often in the isnads than later ones. Because this is the case, there is usually a relatively large discrepancy between the number of isnads, hadiths, and tarafs in which an early relationship occurs. Calculating the amount of hadith transmission activity based on isnad or

hadith count would have made it seem like Jarir b. ʿAbdullah transmitted the five pillars hadith, for example, 320 (based on isnad count) or 214 (based on hadith count) times. While one can imagine that in certain scenarios that this could be true, it is safer to assume that Jarir transmitted each *taraf* once and that the differences in the versions of the same hadith ultimately recorded later in hadith collections containing the **Jarir b. ʿAbdullah/**ID: 2123 → **al-Shaʿbi/**ID: 4099 relationship result from the vagaries of the hadith transmission process rather than as a reflection of multiple instances of transmission between them. Thus, we use the number of times a given transmitter relationship occurs in different *tarafs* as the basis for calculating hadith transmission activity. For the Jarir → Shaʿbi relationship this would be 11 *tarafs*.

In addition to the ambiguity on the location of transmitter relationships in space and time, given the city and year data that we have, there is one other obstacle: we do not have year and city data for all narrators of the hadith transmission network. Out of the 49,819 unique narrators found in the totality of *GK* hadith corpus, 9635 of them have both time and geographical information. For this reason, we built a version of the hadith network that consisted of only these 9635 narrators.

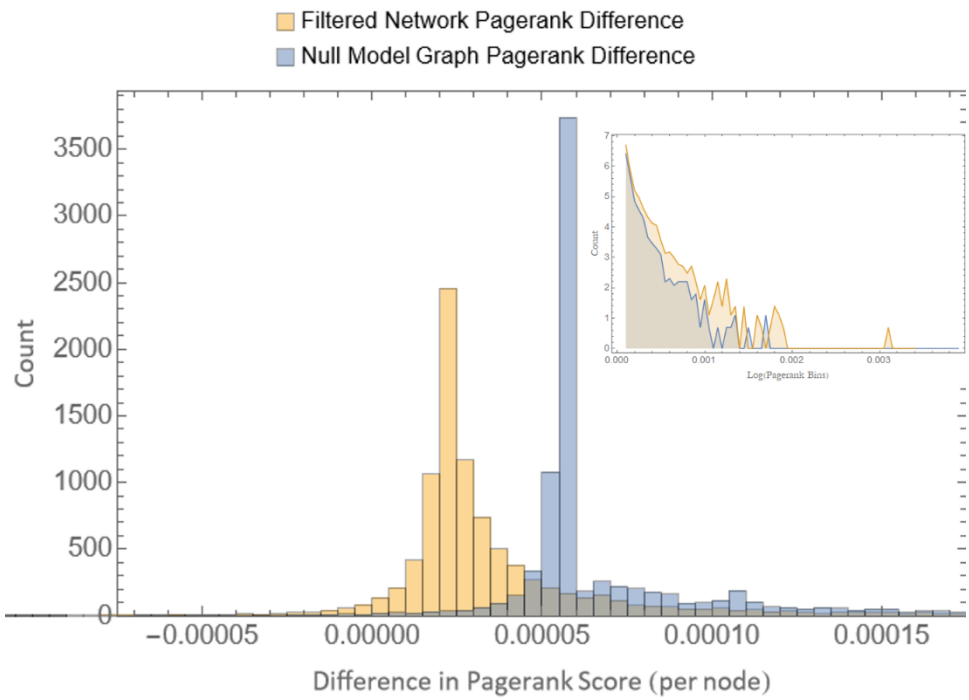

Figure 2. Verification of network structure preservation when sampling transmission edges that have biographical information. Actual pagerank comparison (log(pagerank)) is shown in the inset (blue is the filtered network's pagerank distribution in the inset).

This raises the question of how well the hadith network we constructed, which only includes transmitters that have both city and year information, represents the much larger one with all the narrators GK identified in the isnads. We tested this by transforming the two networks into mathematical objects called graphs, which allows us to compare their relative structures. Graphs at a minimum consist of nodes, which in our case would be narrators, and edges, which represent the transmitter relationships between narrators. One graph ("null model graph") consisted of a random sampling from the set of all reliable nodes and edges (the "full network") found in the GK dataset. Another graph ("filtered network") consisted of only those nodes and edges for which we have city and time information. One way to see how well our filtered network retains the basic structure of the full network is to compare the distribution of the scores that quantify a node's prominence between the same nodes in the two networks. One such metric Pagerank, commonly used by social network analysts, is a mathematically

calculated number for each node in a network that represents its prominence relative to other nodes, based solely on the structure of the relationships between the nodes. To do the comparison, we simply subtract the filtered network's nodes' values from their counterpart in the full network. The same is done for the null model graph. Networks that are structurally similar should have a similar distribution of Pagerank and Pagerank difference scores. Figure 2 represents the distribution of these differences of Pagerank numbers.[10] A median value of 0.0 in these histograms would mean that the networks are exactly the same. As can be seen from the histograms, the median of the difference values (peak of the distribution) is closer to 0.0 for the filtered network, suggesting that its transmission structure mimics the full network significantly more than the null model network. Another way to compare the structures of the filtered network with the full network is by comparing the logarithm of the Pagerank scores of each individual node in both networks (figure 2, inset).[11] The shapes of the distribution are similar, which indicates that both networks have similar structures of relationships between nodes. Finally, although our filtered network extracts 9,635 narrators from a total of the 49,819 (19.34%) found in the overall network, these 9,635 narrators comprise 119,729 (~37%) of the total 320,393 transmitter relationships in the overall network. The fact that ~20% of the narrators still retained ~37% of the relationships is an indication that the filtered network is represents the core of the full network. The computational analysis above coheres with the reasonable qualitative historical supposition that the collectors of biographical information focused on gathering information, especially parameters like birth, death years, and cities of the most prominent narrators. Based on this analysis we concluded that the filtered network we relied on for this paper is a good approximation of GK's full network.

[10] For an introduction to Pagerank, see Monica Bianchini, Marco Gori, and Franco Scarselli, "Inside pagerank," *ACM Transactions on Internet Technology (TOIT)* 5, no. 1 (2005).

[11] Logarithm is especially useful in revealing the large-scale features of the histograms.

Analyzing the Quantity of Transmissions

We applied our algorithms to the network to distribute 119,729 transmitter relationships into 593 unique cities across 884 years (-11 to 872). We can now calculate the amount of transmission activity in any given city for any given year and compare its rise and fall across many cities and over time. Below, we present our analysis of the rise and fall of transmission activity in the four most dominant cities covering the formative and classical periods of the hadith transmission network.

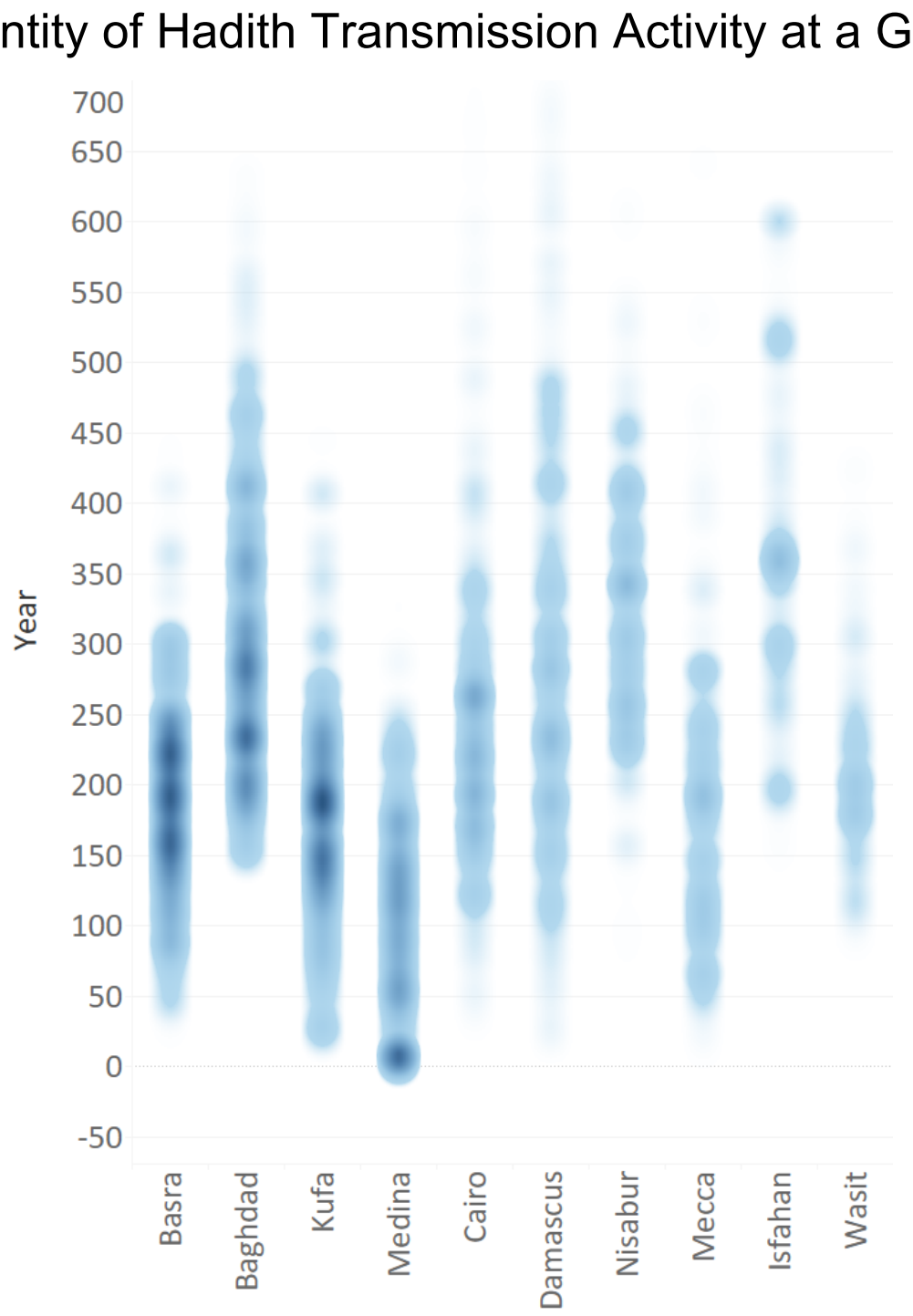

Figure 3. Quantity of transmissions represented as density in top 10 cities over time.

Figure 3 shows a temporal density map of the quantity of hadith transmission in the top ten cities by year. This visualization reveals the years in which each city was active, and the rise

and fall of their transmission activity. Medina, Kufa, Basra and Baghdad, specifically have the highest transmission activity for the time span considered here. We investigate why this is the case.

Correlation of the Rise and Fall of Dominant Cities with Demographic, Political, and Economic Factors

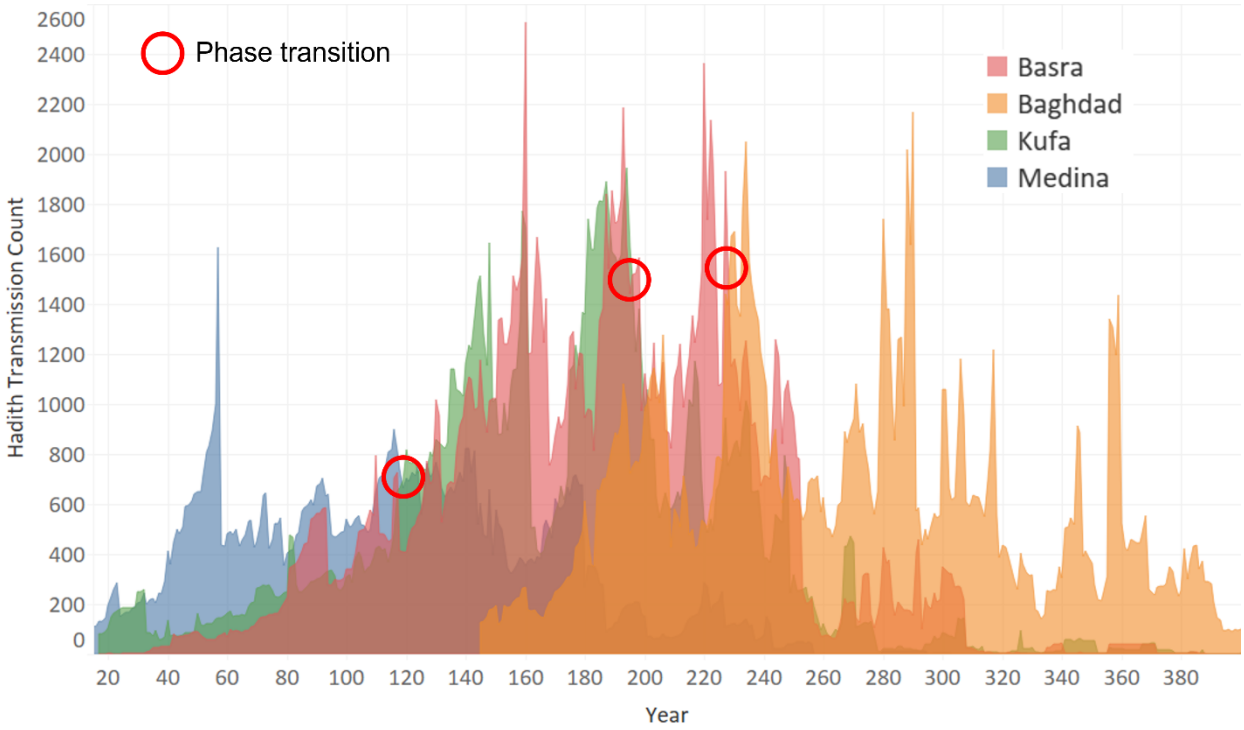

Figure 4. Overall transmission activities in four major cities. The red circles denote when an activity curve surpasses other curves for a sustained time (phase transition).

In figure 4, we identify three phases in which four top cities dominated: Medina recorded the highest hadith transmission count in phase I; Kufa and Basra traded spots in phase II; and Baghdad dominated in phase III.[12]

[12] Since the Islamic calendar begins with the Prophet Muhammad's emigration from Mecca to Medina, we use the negative dating scheme to denote the time before the calendar's start. We choose -11 as the earliest time that hadiths between the Prophet and his companions could have been transmitted because that is the year in which the Prophet began his prophetic mission.

How does these phases compare with what we know of the rise and fall of the demographic, economic, political, and cultural vitality of each of these respective cities? The following sections dive deeper into the dynamics for each phase. To better explain each city's activity curve (as the overlapping shades in figure 5 cover up some activities), we have divided the sections based on phases.

Phase I: Medina's Dominance (0-119)

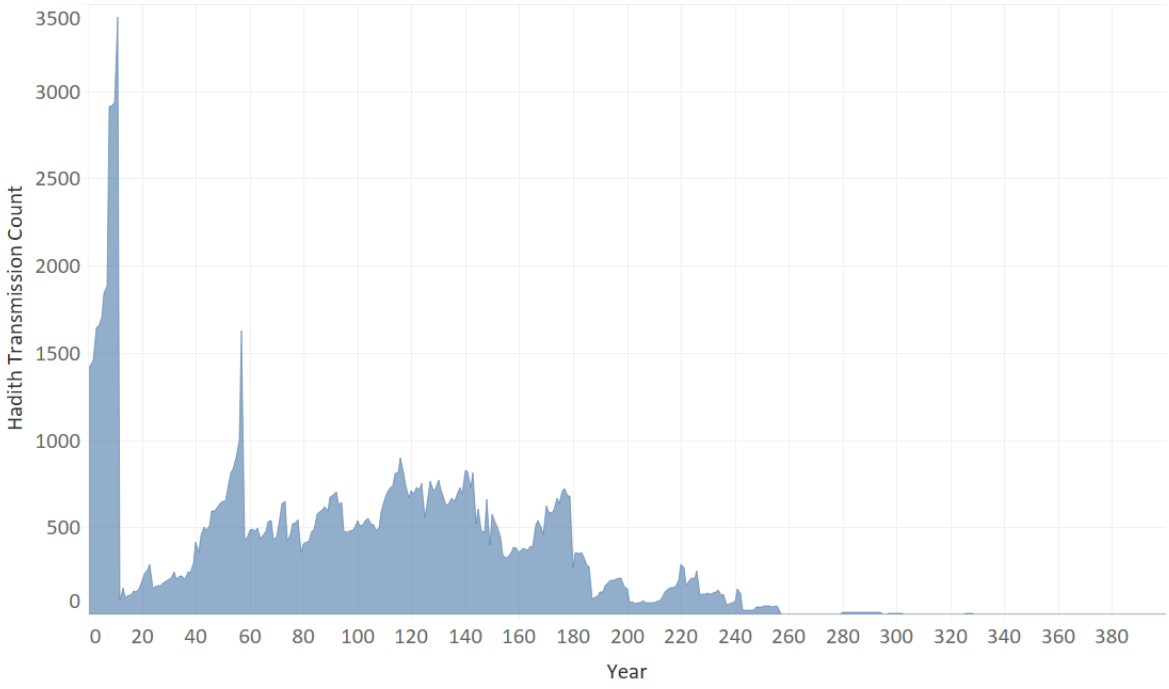

Figure 5. Hadith transmission count in Medina.

With the exception of Medina (figure 5), the respective rise and decline of hadith transmission activity in Kufa, Basra, and Baghdad largely correlate with the changes in the demographic, political, and economic vitality of these cities. Medina, despite dominating hadith transmission activity in this phase, is the exception because it stopped serving as the empire's capital in the year 36, when ʿAli (r. 36-41) became the caliph.[13] However, Medina continued to

[13] W. Montgomery Watt and R.B Winder, "al-Madīna," in *Encyclopaedia of Islam, Second Edition*, ed. P. Bearman, et al., Encyclopaedia of Islam (Leiden: Brill, 1980-2004)

serve as the home of many families that served as the vanguard of the Prophet's message and in the administration of the Umayyad caliphate. In fact, in this phase much hadith transmission originates in a handful of mostly Medinan companions that died in the fifth, sixth, and seventh decades of the first centuries, the most prominent being Abu Hurayra (d. 57), ʿAbdullah b. ʿUmar (d. 73), ʿAʾisha bt. Abi Bakr (d. 57), and ʿAbdullah b. ʿAbbas (d. 68). The subsequent generation, also in Medina, continued to evince strong level of transmission activity. For example, current scholarship identifies the activities of one scholar, the Muhammad b. Shihab al-Zuhri (d. 124), who was active in Medina, as especially foundational to the growing formalization of hadith transmission. Thus, despite the eventual political and economic marginalization of Medina by the turn of the first centuries, and its relatively low population in the time period under consideration, early Islam scholarship identifies it as an extremely influential city in precisely the time period our study identifies it as the center of hadith transmission activity.[14] Our quantitative statistical analysis of hadith transmission corresponds with the

[14] Wheatley concludes that both Mecca and Medina had relatively under-developed urban systems in the earliest period of Islam, and were soon eclipsed by the cities established and conquered outside of Arabia. Paul Wheatley, *The places where men pray together : cities in Islamic lands, seventh through the tenth centuries* (Chicago: University of Chicago Press, 2001), 32. During the time of the Prophet, Watt and Winder put Medina's population at 30,000, and say that in the pre-Islamic period it is reported to have had 200 forts Watt and Winder, "al-Madīna," . ʻAbd al-ʻAzīz Dūrī, *The rise of historical writing among the Arabs*, trans. Lawrence I. Conrad (Princeton, New Jersey: Princeton University Press, 1983), 22 identifies Medina's dominance in the early period of hadith transmission and places it as the origin of hadith-based historiography as well as the source for the insistence on isnad as a source verification method in addition to the place where narrator criticism developed. For an account of the activities of the seven individuals later Sunni tradition identifies as important religious scholars and hadith collectors in the generation of the successors, see Ch. Pellat, "Fuḳahāʾ al-Madīna al-Sabʿa," in *Encyclopaedia of Islam, Second Edition*, ed. P. Bearman, et al., Encyclopaedia of Islam (Leiden: Brill, 1980-2004). For a prosopographical study of elite Hijazi families in early Islam see Asad Q. Ahmed, *The religious elite of the early Islamic Ḥijāz : five prosopographical case studies*, Occasional publications of the Unit for Prosopographical Research ; 14 (Oxford: Prosopographica et Genealogica, 2011). The amount of information recorded about the marriages, divorces and deaths of elite Hijazi families and their involvement in government administration, whether for the Umayyads or Abbasids is an indication for the continued importance of Medina well into the early Abbasid period. For another example of scholarly activity in Madina involving important religious authorities and even Umayyad elites such as ʿAbd al-Malik b. Marwān, see Sean W. Anthony, Muhammad and the empires of faith the making of the prophet of Islam, (Oakland, California: University of California Press, 2020), https://www.jstor.org/stable/10.2307/j.ctvw1d5tc. 86-95.

qualitative historical picture historians have drawn of intellectual activity in Medina for the time period under consideration.

Phase II: Kufa and Basra (120-159); Dominance of Kufa (160-193) and Basra (194-227)

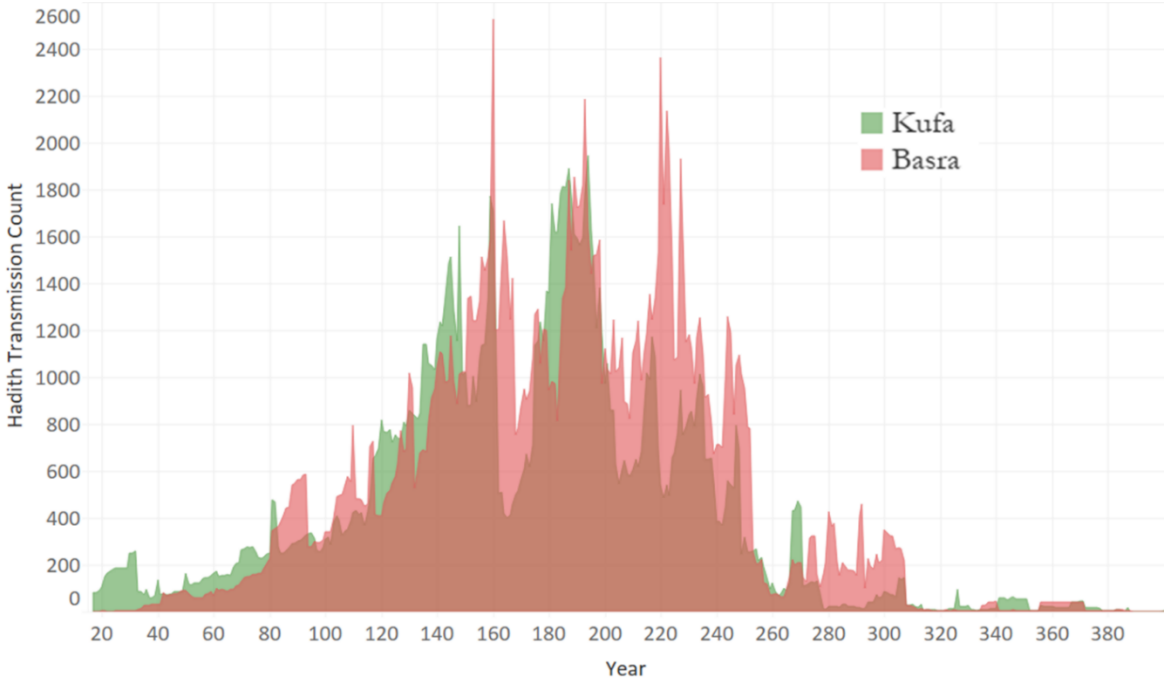

Figure 6. Hadith transmission count in Kufa (green) and Basra (red).

Just as the share of hadith transmission activity began to decline in Medina in the last quarter of the first century, it began to increase first in Kufa, and then subsequently in Basra (figure 6).[15] Along with Medina, by the second half of the first century, the two Iraqi cities of Kufa and Basra had become politically, culturally, and intellectually important to the early Islamic empire. Kufa served as the empire's capital during the albeit short and tumultuous caliphate of ʿAli b. Abi Ṭalib. It was the origin of many ʿAlid rebellions, and competing legal and theological

[15] Lucas locates the center of gravity for what he calls the second phase of hadith transmission in Iraq, more specifically in Kufa and Basra. See Lucas, *Constructive Critics*, 67-68. Jait, more broadly, identifies efflorescence of early Islamic intellectual and cultural activity as first occurring in Kufa, then migrating to Basra, and finally culminating in Baghdad.

communities of early Islam. It served as the center of Abbasid power after the revolution in 132 until al-Mansur's founding of Baghdad in 145. Demographic information about Kufa accords both with the narrative of its political fortunes and the level of hadith transmission that we see over the years. Chronicle sources say that it had a population of 40,000 in 41, which ballooned to 140,000 by 51.[16] Some sources indicate populations between 60,000 men and 80,000 hearths for the middle of the second century. Wheatley puts the second century estimate it at 150,000.[17] Bosker, Buringh, and van Zanden put the number at 100,000.[18] Though we could not find population estimates for subsequent centuries, one historian links it to the establishment of Baghdad in central Iraq, which initially lessened Kufa's political importance and then eventually shifted trade patterns towards Basra as well.[19]

While the founding of Baghdad ultimately weakened Kufa's commercial and cultural importance, its initial impact on Basra, in southern Iraq, had the opposite effect. Given Baghdad's status as the seat of the Abbasid court during much of the dynasty, and its location in central Iraq on the point of confluence between the Tigris and Euphrates, it naturally became a hub for trade. Basra's location near the mouth of Persian/Arabian gulf allowed it to function as an entry point for Indian Ocean trade headed to Baghdad through a maritime route and served as a waystation for overland trade as well. It's legacy as one of the caliphate's (both rightly guided and Umayyad) premiere cities, combined with its newfound economic importance, elevated its cultural prominence.[20] We see this dynamic reflected in the level of hadith transmission activity as well. While Kufa starts to decline about a decade after Baghdad's

[16] The 40,000 number is cited in Hichem Djaït, "al-Kūfa", Encyclopaedia of Islam, Second Edition. Litvak, "KUFA", Encyclopædia Iranica is the source for the 140,000 number.

[17] Found in Wheatley, who cites the number from Baladhuri. Wheatley, *The places where men pray together : cities in Islamic lands, seventh through the tenth centuries*, 49.

[18] Maarten Bosker, Eltjo Buringh, and Jan Luiten van Zanden, "From Baghdad to London: Unraveling Urban Development in Europe, the Middle East, and North Africa, 800–1800," *The Review of Economics and Statistics* 95, no. 4 (2012): Table A2

[19] Litvak, "KUFA", Encyclopædia Iranica; F. M. Donner, "BASRA", Encyclopaedia Iranica.

[20] F. M. Donner, "BASRA", Encyclopaedia Iranica.

founding, Basra flourishes as the center until the first quarter of the third century, when Baghdad replaces it as the most important city for hadith transmission. This correlates with the demographic decline of Basra, the cause of which may be related to the region's political instability in the fourth century, specifically attacks on it by surrounding tribes.[21]

The available demographic data also correlates with the rise and decline of transmission activity. The estimates point to the fact that Kufa and Basra had very similar demographic profiles in the first century, with Basra having a bit of a larger population. By the end of the first century, estimates place the population at 200,000.[22] In the second and third centuries estimates of Basra's population range between 200,000 and 600,000.[23]

Phase III: Baghdad's Dominance (228-400)

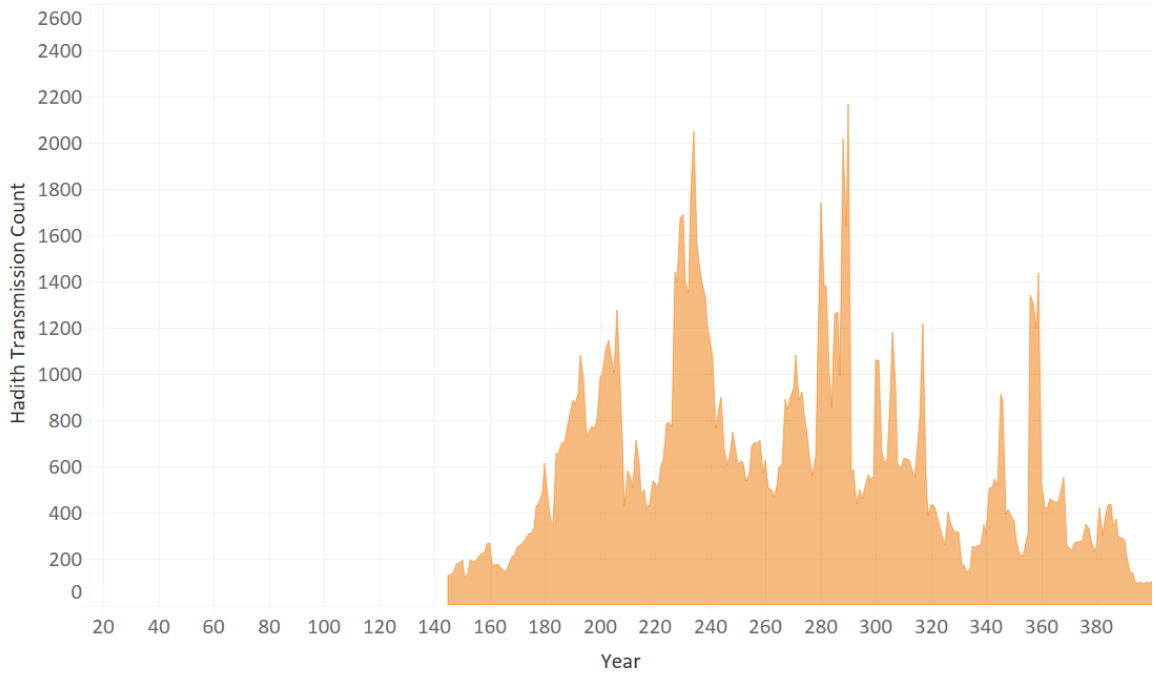

Figure 7. Hadith transmission count in Baghdad.

[21] Ibid.

[22] Wheatley gives 80,000 as the number of individuals recorded in Basra's diwan Wheatley, *The places where men pray together : cities in Islamic lands, seventh through the tenth centuries*, 45 in the year 50. He cites Pellat as estimating the population to be 200,000 in the year 100.

[23] Pellat Charles and H. Lang Katherine, "Basra until the Mongol conquest", Encyclopaedia of Islam, THREE.

Historians say Baghdad became the most important center for intellectual activity generally and the most important center of hadith transmission in the third century.[24] Our data confirms these claims (figure 7), and at least for transmission activity gives more precise time-bounds. The rise and eventual fall of hadith transmission activity correlates with the rise and fall of Baghdad's population, reaching 350,000 in the third century, growing to 450,000 in the beginning of the fourth, climbing to 1.5 million by the end of it.[25] In addition to correlating with changes in population, the emergence of Baghdad as a center for hadith transmission also tracked its rise as a political center and hub of economic activity. Baghdad served as the capital of the Abbasid empire on and off through the second century, and remained as its economic and cultural center throughout the third.

[24] Gregor Schoeler, *The Oral and the Written in Early Islam*, ed. James E. Montgomery, trans. Uwe Vagelpohl (New York: Routledge, 2006), 116. Schoeler makes the point to note that when Baghdad eclipsed Kufa and Basra as the center for hadith transmission the insistence on oral transmission of hadith in formal settings also eventually disappeared. For Baghdad's emergence as the center in the third century, see Lucas, *Constructive Critics*, 86-88. Schoeler does not provide a source for this claim; one senses it is an intuition. Lucas, on the other hand, relies on it based on a chronological framework that divides the hadith transmission into seven distinct phases, while identifying the most prominent narrators in each phase, and noting the cities in which they lived. Djaït identifies Baghdad as the heir to intellectual and cultural fermentation that took place in Kufa and Basra in the first/seventh and second/eighth centuries Djaït, "al-Kūfa", Encyclopaedia of Islam, Second Edition.

[25] At its founding in 145, Baghdad had, at a minimum, a population of 100,000, because chronicles record that as the number of workers needed to build it A.A. Duri, "Baghdād", ibid. For the estimate of 350,000 in the beginning of the third and 450,000 in the beginning of the fourth century, see Bosker, Buringh, and van Zanden, "From Baghdad to London: Unraveling Urban Development in Europe, the Middle East, and North Africa, 800–1800,". For the estimate of 1.5 million at the end of the fourth (more precisely 383/993), see Duri, "Baghdād", Encyclopaedia of Islam, Second Edition.

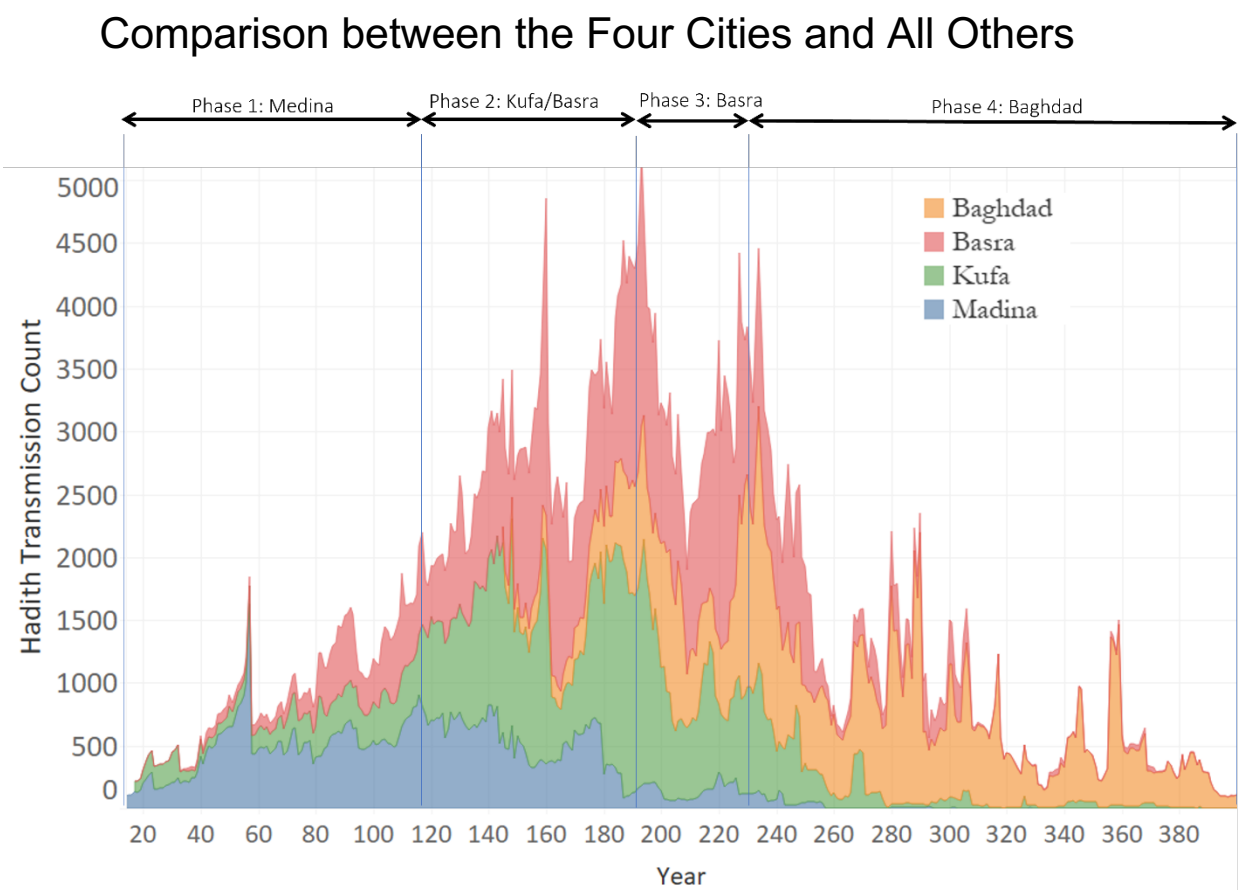

Figure 8. Stacked area plot for all cities.

Figure 8 shows a stacked area plot, where the quantity of hadith transmission for the four cities are stacked for each year, such that the relative area (as opposed to the y-axis ticks) indicates the quantity for a given city. The initial years of activity in Medina are left out to allow a more detailed zoomed-in view of later years. Transmission activity, in terms of absolute numbers, is the highest in phase II and III. In comparison to figure 4, the counts are less. However, the overall shape and peaks of the curves (in figures 4 and 9) are similar, indicating the dominance of these four cities further.

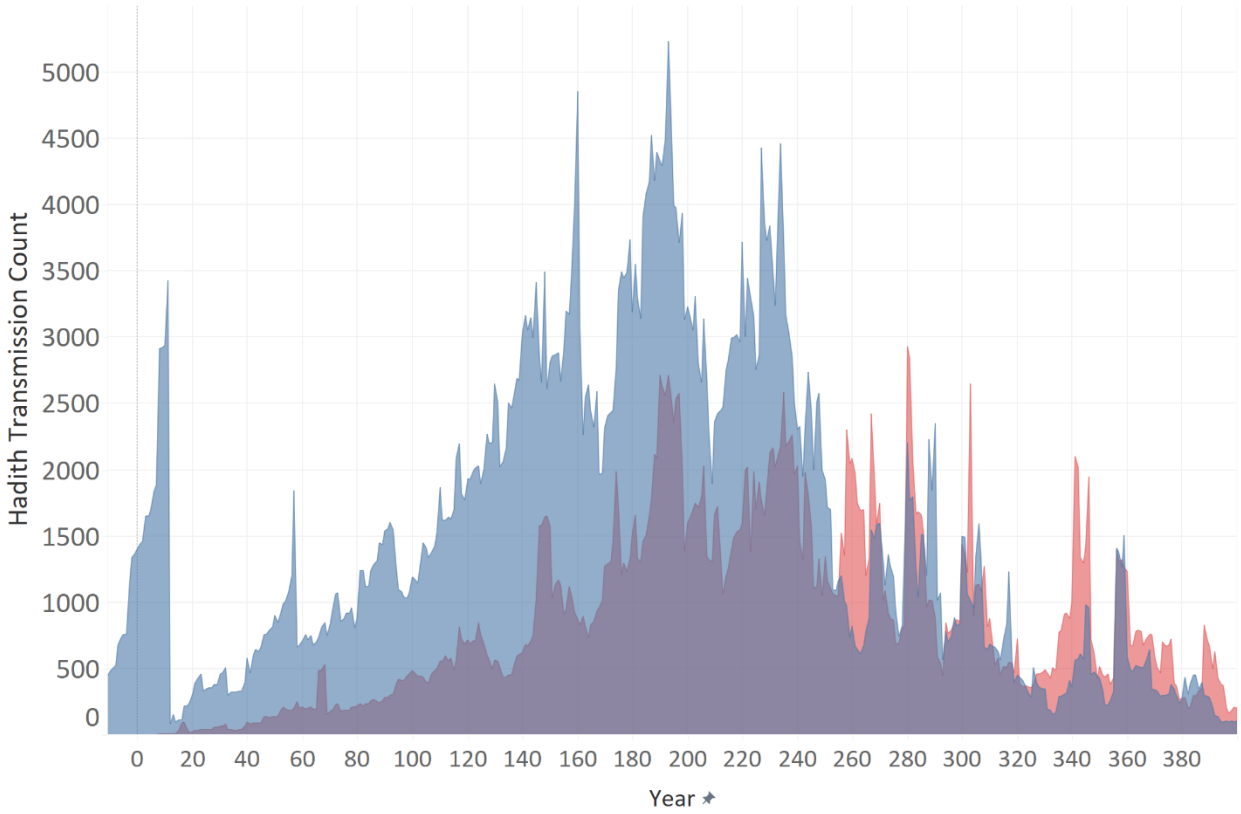

Figure 9. Transmission activity in dominant four cities vs. all others.

Figure 9 provides a sense of the dominance of the four cities compared to all 589 other cities for the period under consideration. One notices two things: the extent to which the total transmission activity in any given year is accounted for by the four major cities under consideration and a noticeably decline of their dominance starting around the year 300. Further analysis of these dynamics and the structure of the distribution of transmission activity across different cities requires investigation that is beyond the purview of this paper.

Conclusions and Implications

The fact that the rise and fall of the dominant cities for hadith transmission activity correlates with their political, economic, and demographic fortunes is broad indication of the fidelity of the bulk of the transmitter relationships that constitute the hadith transmission network.

Thus, the total rejection of the isnad as a citation system that conveys historical information, an influential position within Western hadith studies, is unjustified. Our findings suggest that scholars of early Islam should not simply assume that isnads are fabrications. With that said, an acceptance of the potential historicity of isnads at a global level does not mean that individual hadiths, or even sets of hadiths, or certain parts of the isnads in the hadith literature are not spurious. Rather, the dating and authentication of individual hadiths requires the application of methods that involve close textual analysis.

In addition to the implication of our findings for existing debates on the historicity of isnads, our paper also introduces a new object of study in its own right: the hadith transmission network. By supplementing biographical data for the narrators with the vast trove of transmitter relationships data found in isnads, we enriched and extended Senturk's pioneering conceptualization. By developing a method to extend the discrete year and city data associated with 9635 individual narrators to ~120,000 transmitter relationships, we created a dataset that enables scholars to pursue answers to old and new questions. For example, we can now precisely quantify the movement of hadiths from one city to the next over time. In addition, up until now, tracing the circulation of hadith texts in time and space through the analysis of their isnads was a laborious process. Our work, to a large extent has automated that process by placing *GK*'s *taraf* IDs in different cities and years. At a minimum, our dataset can provide an initial automated first-pass placement of hadiths in cities and years that scholars can subsequently modify, accept, or reject based on traditional close-reading techniques of textual analysis.[26]

We extended the conceptualization of the hadith transmission network not only by combining narrator biographical data with information found in isnads, but also by treating it as a

[26] For a potential use case of how dating and pinning the geography of hadith can help in the history of Islamic law see, see Najam Haider to Islamic Law Blog, 2020, https://islamiclaw.blog/2020/12/22/future-avenues-in-the-study-of-islamic-law/

formal mathematical object – a graph. A graph, which at core simply consists of a set of nodes and relationships between them, is capable of conveying very interesting statistical information about the features of the network it represents. Social scientists have been using the mathematical properties of graphs to analyze social networks for decades. We incorporated the hadith network's graph properties in two ways. First, we used different versions of the hadith transmission graph's distribution of Pagerank scores to make claims about how the filtered city-year network retains structural features of the overall hadith network. Second, we used the network's graph properties to quantify the extent to which a given narrator belonged to multiple cities based on his immediate transmitter relationships (see Appendix 1). This only scratches the surface of how computational and mathematical methods of social network analysis can be used to analyze the hadith transmission network. Future research can apply and interpret computational techniques to determine a narrator's prominence, or the existence of distinct communities implied simply in the structure of the relationships different narrators have to each other.

Finally, given the sheer temporal longevity and the relatively large number of premodern individuals involved in the transmission of information, historians have yet to adequately characterize and assess the social, political, and economic effects of the hadith transmission network's existence.[27] To what extent can its existence, the density of human relationships it contained, and its perdurance through the centuries explain the phenomenal success of Islamic cultural values and civilizational patterns in premodern history?

[27] Only recently have scholars begun looking systematically at hadith transmission practice outside of the formative and classical periods. See, importantly, Garrett A. Davidson, *Carrying on the tradition: a social and intellectual history of Hadith transmission across a thousand years* (Leiden: Brill, 2020).

Appendix 1: Detailed Description of the City and Year Placement Algorithm

Extending City Assignments in Narrator Biographies to Transmitter Relationships

We use the city information found in the biographies of the narrators to ultimately place a transmission in space, and we use the instances of transmissions between narrators found in the isnads of hadith texts to provide the amount of transmission activity. The central challenge is how we extend city information, which GK assigns to individual narrators, to transmitter relationships, which consist of two narrators. Because GK often assigned individual narrators to multiple cities, it is often unclear where to place the relationship.[28] Specifically, there are three possible cases for any given relationship: first, both transmitters share a city; second, the transmitters do not share any cities; and third, the transmitters share multiple cities.

For the first case, even if both transmitters belong to multiple cities, as long as they share a city, we deem this the place of transmission. For example, multiple isnads of the five pillars hadith cite the following transmitter relationship: **Israʾil b. Yunus**/ID: 982 → **Ishaq b. Mansur**/ID: 969, a relationship occurs in 43 other hadiths as well. *GK* assigned Israʾil to Baghdad and Kufa and Ishaq to only Kufa. The single shared city between them is Kufa, and is logically the most probable city in which they transmitted hadith, therefore, we assign the 44 hadiths in which this relationship is cited wholly to Kufa.

The second case involves the scenario where the source and recipient in a transmitter relationship do not share any cities. In this case, we place the transmission in the source's city or cities, on the assumption that it would have been more likely that the typically younger receiver would have travelled to the typically older source's place to hear his hadith. For

[28] GK assigned ~56.7% (5464 of 9635) one city, and the rest to more than one. GK assigned the vast majority of the remaining narrators 2-4 cities, with the maximum of 21, and a mean of ~1.77.

example, If *GK* assigned the source to a single city, then we placed the transmission wholly in that city.

The last case is the most complex: when the source and recipient share multiple cities. Instead of arbitrarily picking just one of the common cities, we distribute the transmissions into all of them based on ratios representing the extent to which the source belonged to the multiple cities he shares with the recipient. We explain how we calculated these ratios below.

Calculating Extent of a Narrator's Belonging to Multiple Cities

If a transmitter is assigned multiple cities, we can get a sense of which cities he is *mainly* from by looking at cities assigned to his sources and recipients. That is, if all his acquaintances are from a particular city, we can sensibly say that the transmitter is also more from that city than others. Inspired by this idea, we developed a ratio for each city assigned to him by GK, by looking at the cities of all his immediate narrators, both those that he transmitted hadiths to, and those he received hadiths from. These proportions quantify the extent to which the narrator belongs to the city *GK* assigned him. Let's take a look at the procedure through the example of the **Ruh b. ʿUbada**/ID: 2969. Overall Ruh is involved in 282 unique transmitter relationships found in the texts of many different hadiths. He was the transmitter to 86 other narrators and a recipient from 195. Overall, these 282 unique transmitter relationships were found in 3,701 hadiths. Since we are only interested in using the city ascriptions of Ruh's transmitters and receivers to determine the proportional extent to which he belongs to his own two cities, Baghdad and Basra, we are uninterested in those narrators that belong to cities other than Baghdad and Basra. This leaves us with 187 narrators that either received from or transmitted hadiths to him and were assigned Baghdad or Basra (or both). One way to get a ratio which represents the extent to which Ruh belonged to Baghdad and Basra would be simply to add up all of the narrators who belonged to Basra and Baghdad respectively and then divide each

number by the total number of Baghdad and Basra assignments found in the 187 narrators. This method assumes that each transmitter relationship between Ruh and other narrators is equal. But this is not the case. For example, Ruh transmits 518 hadiths to Ahmad b. Hanbal (d. 241), his most prolific receiver. We decided to take this into account by strengthening transmitter relationships in proportion to the number of hadiths received or transmitted in that relationship. The result of this operation for Ruh is assigning him to Baghdad at a 0.534275 ratio and Basra at 0.46725. How do we then use these ratios? Consider the **Ruh b. ʿUbada**/ID: 2969 → **Muhammad b. Yahya**/ID: 7323 transmitter relationship, which is cited in 9 hadiths. Ruh was assigned to both **Baghdad** and **Basra**, and Muhammad b. Yahya was assigned to Nisabur, **Baghdad** and **Basra**. Thus, they share two cities: **Baghdad** and **Basra.** Since Ruh is the source in this transmitter relationship, and generally speaking recipients travelled to the cities in which prominent sources were located in order to hear their hadith, we use the city ratios we developed for him to determine the distributing of the 9 hadiths to Baghdad and Basra: 4.808475 (0.534275*9) of the transmissions are placed in Baghdad and the rest, 4.20525 (0.46725*9), are placed in Basra.

Extending Narrator Birth and Death Years to Span of Transmitter Relationship

Just as we make inferences on *where* transmission took place given the data we have at hand, similarly we make inferences on *when* transmission took place. The temporal data associated with the narrators is much more predictably structured than the geographical data. Whereas there was quite a bit of variation in the number of cities *GK* assigned a narrator, this is not so much the case with time. To place the transmissions in a timespan, we created a model predicated on the birth and death years of the teacher and student. Because we only selected narrators found in isnads, if, at a minimum, they had both at least a death year and one city, all of our narrators have death years but not all of them have birth years. Out of 9,635 narrators,

we have birth years for 2,925 of them. To provide birth years for the rest, we calculated the median life span of a transmitter in our corpus, which is 80 years, and subtracted it from their death year.

Once we have birth and death years for both narrators of a transmitter relationship, we can calculate the overlap of their lifespans (indicated by the orange bar in figure 11). But we assume that the transmission does not take place when either the source or the recipient is a child (indicated by the dashed bars in figure 11). Based on Aʿzami's finding that transmitters typically did not begin hearing hadith before the age of 20, we say that the transmission between narrators likely did not take place in the first 20 years of their life.[29] Thus, in those cases where the overlap between narrators was more than 20 years, we place the transmissions in the years of the overlap only after both have passed the age of 20. If this not possible the overlap between the transmitters' lifespans was less than 20 years, we simply place the transmissions in the duration of the overlap.

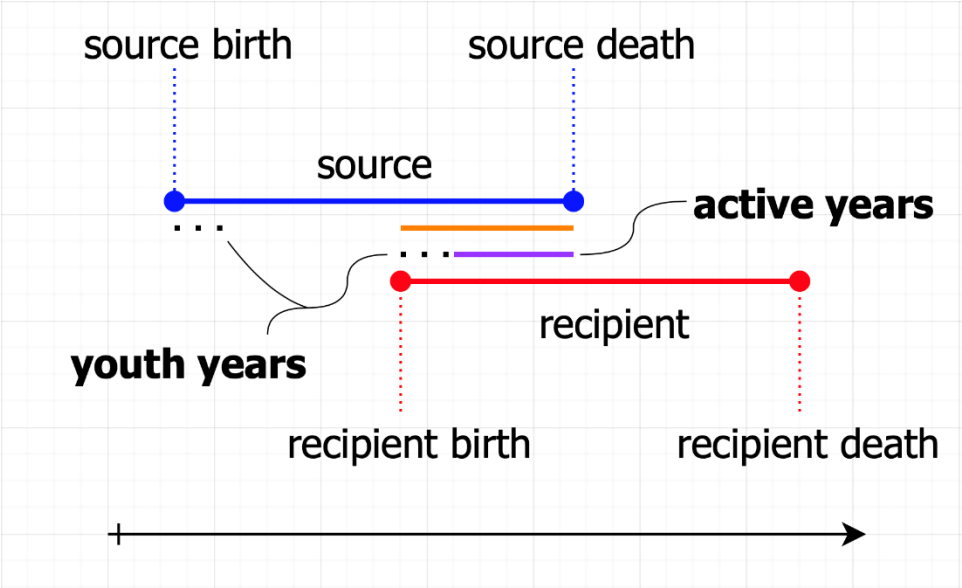

Figure 10. Teacher/Student transmission overlap.

[29] Al-Azami, based on anecdotes found in works of hadith criticism and terminology, holds that students started joining the circles of the hadith scholars at 20 years of age. See al-Azami Muhammad Mustafa, On Schacht's origins of Muhammadan jurisprudence, (Riyadh: King Saud University, 1985), https://library.dctabudhabi.ae/sirsi/detail/151225. 197-98

As an example, let us consider again the Ruh →Muhammad transmitter relationship. We have the following three data points: Ruh's death year is 205, Muhammad's birth year is 172, and his death year is 258.[30] Since we do not have Ruh's birth year, we use the median life-span (i.e. 80 years), and assume that Ruh was born in the year 125 (205 minus 80). Thus, their life spans overlapped for 33 years (Muhammad's birth year through Ruh's death year (172-205)). While it is physically possible for Muhammad to have "heard" Ruh's hadiths at any point during their overlap, it is unlikely that this would happen when Muhammad was a child. Therefore, the actual active span in which the transmission could have taken place from Ruh to Muhammad is in the last 14 years of Ruh's life, namely in the years 192 to 205.

With no way of weighting each of the 14 years differently in a non-arbitrary way, we assumed that the possibility of Ruh transmitting his hadiths to Muhammad was equal for each year, which in this case would be 1/14 or ~7.14%. We can now put together our inferential assignment of the Ruh→Muhammad nine transmissions to their cities and their years (seen in table 3).

Transmission Distribution		
Count:		9 hadiths
Span:		192 to 205 (14 years)
City	City Ratio	Quantity Per Year
Basra	0.4678	9*(1/14)*0.4678 = 0.301 hadiths
Baghdad	0.5322	9*(1/14)*0.5322 = 0.342 hadiths

Table 3. City and Year Transmission Distribution Example

[30] GK does not record a birth year for Rūḥ.

Appendix 2: *GK*'s Sources for Biographical Data and Its Reliability

The information that most difficult to systematically glean from transmitter biographies are the cities where narrators were recorded to have lived or transmitted hadith during their life. *GK* provides a general account of the types of sources that they consulted to create the table containing biographical information.[31] In addition to biographical dictionaries, it is also clear that they made comprehensive use of dozens of works of narrator criticism, which often also contain geographical and temporal information on narrator criticism. We know this because in addition to basic biographical information on narrators, they also provide a summary of narrator critic evaluations of the narrator in question. Unfortunately, while *GK* provided a general account of the types of sources that they relied on to collect biographical information on the narrators and disambiguate the occurrences of names in the isnads, they did not provide the specific sources that they relied on for each individual narrator. The one exception to this is *GK*'s inclusion of the entirety of the biographical entry of a specific narrator from Mizzi's (654/1256-742/1341) *Tahdhib al-Kamal* presumably when such an entry for the narrator was present in Mizzi's collection. Of the 9,654 narrators that make up our dataset of narrators, *GK* has included Mizzi's biographical entries for 1,366 of them.

We used the digitized Mizzi biographies to test the reliability of GK's city ascriptions. We automatically searched for the names of cities *GK* had recorded for each of the 1,366 narrators in the digitized text of Mizzi's biographical entry. For 1,135 of these narrators, every city that *GK* ascribed to a narrator was found in the Mizzi biographical entry. For 37 narrators, none of the places *GK* ascribed to a narrator were found in their respective Mizzi entries. For 183, over half of the cities *GK* ascribed to the narrators were found in the Mizzi entries, and 9 entries had less

[31] For a description of the sources *GK* consulted for the biographical information, see section 5, *khidamat ruwat al-ḥadith*, especially its subsection *manhaj al-ʿamal fi tarajim al-ruwat*. This document is found in the desktop version of their software and may be accessed through the *taʿrifat* tab.

than half. Based on this data, we concluded, *GK* proved reliable in mining city ascription data from a source heavily relied upon for biographical information on narrators.[32] Even for the statistically handful that they did not either find all or part of the cities they ascribed to the narrators, it is quite possible that they consulted other biographical sources that ascribed those cities to the narrators.

[32] Juynboll calls Mizzi's *Tahdhib al-kamal fi asmaʾ al-rijal*, "a milestone in the ʿilm al-rid̲j̲āl in that it is the first comprehensive lexicon that aims at being exhaustive, much more so than any of its predecessors." G.H.A. Juynboll, "al- Mizzī", Encyclopaedia of Islam, Second Edition.

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